测地线方程:

$$\begin{cases} \ddot{u} + \Gamma_{11}^1 \dot{u}^2 + 2\Gamma_{12}^1 \dot{u}\dot{v} + \Gamma_{22}^1 \dot{v}^2 = 0 \\ \ddot{v} + \Gamma_{11}^2 \dot{u}^2 + 2\Gamma_{12}^2 \dot{u}\dot{v} + \Gamma_{22}^2 \dot{v}^2 = 0 \end{cases}$$

$\dot{r} = \dot{u} r_u + \dot{v} r_v$ 协变导数:

$$\frac{D}{dt}\dot{r} = (\ddot{u} + \Gamma_{11}^1 \dot{u}^2 + 2\Gamma_{12}^1 \dot{u}\dot{v} + \Gamma_{22}^1 \dot{v}^2) r_u + (\ddot{v} + \Gamma_{11}^2 \dot{u}^2 + 2\Gamma_{12}^2 \dot{u}\dot{v} + \Gamma_{22}^2 \dot{v}^2) r_v$$

1. $r(u,v) = (a\sin u\cos v,\ a\sin u\sin v,\ a\cos u)$ $\quad u \in [0,\pi)$ $\quad v \in [0,2\pi)$

$r_u(u,v) = (a\cos u\cos v,\ a\cos u\sin v,\ -a\sin u)$

$r_v(u,v) = (-a\sin u\sin v,\ a\sin u\cos v,\ 0)$

$E = r_u \cdot r_u = (a\cos u\cos v,\ a\cos u\sin v,\ -a\sin u)(a\cos u\cos v,\ a\cos u\sin v,\ -a\sin u)$

$\quad = a^2$

$F = r_u \cdot r_v = -a^2\cos u\sin u\cos v\sin v + a^2\cos u\sin u\cos v\sin v = 0$

$G = r_v \cdot r_v = a^2\sin^2 u\sin^2 v + a^2\sin^2 u\cos^2 v = a^2\sin^2 u$

$EG - F^2 = a^4\sin^2 u$

$$\Gamma_{11}^1 = \frac{GE_u - 2FF_u + FE_v}{2(EG-F^2)} = 0$$

$$\Gamma_{12}^1 = \frac{GE_v - FG_u}{2(EG-F^2)} = 0$$

$$\Gamma_{22}^1 = \frac{2GF_v - GG_u - FG_v}{2(EG-F^2)} = \frac{-a^2\sin^2 u\ 2a^2\sin u\cos u}{2a^4\sin^2 u} = -\sin u\cos u$$

$$\Gamma_{11}^2 = \frac{2EF_u - EE_v - FE_v}{2(EG-F^2)} = 0$$

$$\Gamma_{12}^2 = \frac{\cos u}{\sin u}$$

$$\Gamma_{22}^2 = \frac{EG_v - 2FF_v + FG_u}{2(EG-F^2)} = 0$$

$$\begin{cases} \ddot{u} - \sin u \cos u \, \dot{v}^2 = 0 \\ \ddot{v} + \dfrac{2\cos u}{\sin u} \dot{u}\dot{v} = 0 \end{cases}$$

$$\frac{D}{dt}\dot{r} = (\ddot{u} - \sin u \cos u \, \dot{v}^2) r_u + (\ddot{v} + \frac{2\cos u}{\sin u} \dot{u}\dot{v}) r_v$$

2. $r(u,v) = (a\sinh u \cos v, \ a\sinh u \sin v, \ b\cosh u)$ $\quad u \in [0, +\infty) \quad v \in [0, 2\pi)$

$$\boxed{\begin{aligned} &(\cosh u)' = \sinh u \\ &(\sinh u)' = \cosh u \\ &\cosh^2 u - \sinh^2 u = 1. \end{aligned}}$$

$r_u(u,v) = (a\cosh u \cos v, \ a\cosh u \sin v, \ b\sinh u)$

$r_v(u,v) = (-a\sinh u \sin v, \ a\sinh u \cos v, \ 0)$

$$E = r_u \cdot r_u = a^2\cosh^2 u \cos^2 v + a^2 \cosh^2 u \sin^2 v + b^2 \sinh^2 u$$
$$= a^2\cosh^2 u + b^2 \sinh^2 u$$

$$E_u = 2a^2 \cosh u \sinh u + 2b^2 \sinh u \cosh u = (2a^2 + 2b^2) \sinh u \cosh u$$

$$F = r_u \cdot r_v = -a^2 \cosh u \sinh u \cos v \sin v + a^2 \cosh u \sinh u \cos v \sin v$$
$$= 0$$

$$G = r_v \cdot r_v = a^2 \sinh^2 u \sin^2 v + a^2 \sinh^2 u \cos^2 v = a^2 \sinh^2 u$$

$$G_u = 2a^2 \sinh u \cosh u$$

$$EG - F^2 = (a^2 \cosh^2 u + b^2 \sinh^2 u) a^2 \sinh^2 u$$

$$\Gamma^1_{11} = \frac{G E_u - 2F F_u + F E_v}{2(EG - F^2)} = \frac{a^2 \sinh^2 u \, (2a^2 + 2b^2) \sinh u \cosh u}{2(a^2\cosh^2 u + b^2 \sinh^2 u) a^2 \sinh^2 u} = \frac{(a^2 + b^2) \sinh u \cosh u}{a^2 \cosh^2 u + b^2 \sinh^2 u}$$

$$\Gamma^1_{12} = \frac{G E_v - F G_u}{2(EG - F^2)} = 0$$

$$\Gamma^1_{22} = \frac{2G F_v - G G_u - F G_v}{2(EG - F^2)} = \frac{-a^2 \sinh^2 u \, 2a^2 \sinh u \cosh u}{2a^2 \sinh^2 u (a^2 \cosh^2 u + b^2 \sinh^2 u)} = -\frac{a^2 \sinh u \cosh u}{a^2 \cosh^2 u + b^2 \sinh^2 u}$$

$$\Gamma^2_{11} = \frac{2E F_u - E E_v - F E_u}{2(EG - F^2)} = 0$$

$$\Gamma^2_{12} = \frac{E G_u - F E_v}{2(EG - F^2)} = \frac{(a^2 \cosh^2 u + b^2 \sinh^2 u) 2a^2 \sinh u \cosh u}{2(a^2 \cosh^2 u + b^2 \sinh^2 u) a^2 \sinh^2 u} = \frac{\cosh u}{\sinh u}$$

$$\Gamma_{22}^{2} = \frac{EG_v - 2FF_v + FG_u}{2(EG-F^2)} = 0$$

测地线方程:

$$\begin{cases} \ddot{u} + \dfrac{(a^2+b^2)\sinh u\cosh u}{a^2\cosh^2 u + b^2\sinh^2 u}\dot{u}^2 - \dfrac{a^2\sinh u\cosh u}{a^2\cosh^2 u + b^2\sinh^2 u}\dot{v}^2 = 0 \\ \ddot{v} + 2\dfrac{\cosh u}{\sinh u}\dot{u}\dot{v} = 0 \end{cases}$$

协变导数

$$\dot{r} = \dot{u} r_u + \dot{v} r_v$$

$$\frac{D\dot{r}}{dt} = \left(\ddot{u} + \Gamma_{11}^{1}\dot{u}^2 + 2\Gamma_{12}^{1}\dot{u}\dot{v} + \Gamma_{22}^{1}\dot{v}^2\right) r_u + \left(\ddot{v} + \Gamma_{11}^{2}\dot{u}^2 + 2\Gamma_{12}^{2}\dot{u}\dot{v} + \Gamma_{22}^{2}\dot{v}^2\right) r_v$$

$$= \left(\ddot{u} + \frac{(a^2+b^2)\sinh u\cosh u}{a^2\cosh^2 u + b^2\sinh^2 u}\dot{u}^2 - \frac{a^2\sinh u\cosh u}{a^2\cosh^2 u + b^2\sinh^2 u}\dot{v}^2\right) r_u$$

$$+ \left(\ddot{v} + 2\frac{\cosh u}{\sinh u}\dot{u}\dot{v}\right) r_v$$